482 [Just four months . . . tiny naked singularity.] Hawking's evidence was published in Hawking (1992a).

CHAPTER 14

483 General comment about Chapter 14: The historical aspects of this chapter are based almost entirely on my own experiences as a participant.

485–486 [Wormholes are not mere figments . . . in 1916,] Ludwig Flamm (1916) discovered that, with an appropriate choice of topology, the Schwarzschild (1916a) solution of Einstein's equation describes an empty, spherical wormhole.

487 Figure 14.2: Kruskal (1960).

490 [We wrote slowly . . . *American Journal of Physics*,] Morris and Thorne (1988).

490 [(the topic of the Hawking and Ellis book)] Hawking and Ellis (1973).

491 [*vacuum fluctuations near a hole's horizon are exotic*:] Hawking inferred this only very indirectly and somewhat tentatively from his discovery of black-hole evaporation. It was firmly demonstrated to be so six years later, by Candelas (1980).

492 [The answer has not come easily,. . . thereby makes them exotic.] See Wald and Yurtsever (1991) and other references cited therein.

494 [In 1955, John Wheeler,. . . quantum foam—] Wheeler (1955, 1957, 1960).

497 [In 1966, Robert Geroch . . . to travel backward in time,] Geroch (1967). Friedman, Papastamatiou, Parker, and Zhang (1988) have given an explicit example of wormhole creation of the sort envisioned by Geroch's theorem.

499 Footnote 8: van Stockum (1937), Gödel (1949), Tipler (1976).

508 [Our paper was published,] Morris, Thorne, and Yurtsever (1988).

509 [we conjectured so in our paper.] Morris, Thorne, and Yurtsever (1988).

509 Footnote 12: Friedman and Morris (1991).

511 [Echeverria and Klinkhammer. . . *two* such trajectories.] Echeverria, Klinkhamnler, and Thorne (1991).

513–514 Box 14.2: Echeverria, Klinkhammer, and Thorne (1991).

513 [Robert Forward . . . discovered a third trajectory] Forward (1992).

515 [but there seem not to be . . . unresolvable paradox.] For a careful and fairly thorough technical discussion of the issue of paradoxes when one has a wormhole-based time machine, see Friedman et al. (1990).

516 [*California* magazine, . . . on Palomar Mountain.] Hall (1989).

517 [Though we were helped . . . Konkowski] Hiscock and Konkowski (1982).

520 [a similar calculation by Valery Frolov . . . our results] Frolov (1991).

521 [we managed to change . . . got published] Kim and Thorne (1991).

521 [*the chronology protection conjecture*,] Hawking (1992b).

521 Footnote 14: Gott (1991).

521 [I am *not* willing to take . . . the laws of quantum gravity.] For a somewhat technical description of my 1993 reasons for skepticism about time machines, and a detailed overview of research on time machines up to spring 1993, see Thorne (1993)